

HTML CODE

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Real-Time Big Data Ingestion Pipeline</title>

  <style>

    * {

      margin: 0;

      padding: 0;

      box-sizing: border-box;

    }

    :root {

      --primary-color: #2563eb;

      --secondary-color: #1e40af;

      --success-color: #10b981;

      --warning-color: #f59e0b;

      --danger-color: #ef4444;

      --dark-bg: #0f172a;

      --card-bg: #1e293b;

      --text-primary: #f1f5f9;

      --text-secondary: #94a3b8;

      --border-color: #334155;

    }

    body {
```

```
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%);

color: var(--text-primary);

min-height: 100vh;

}
```

```
/* Header */
```

```
.header {

  background: rgba(30, 41, 59, 0.95);

  backdrop-filter: blur(10px);

  padding: 1rem 2rem;

  border-bottom: 1px solid var(--border-color);

  position: sticky;

  top: 0;

  z-index: 100;

}
```

```
.header-content {

  max-width: 1400px;

  margin: 0 auto;

  display: flex;

  justify-content: space-between;

  align-items: center;

}
```

```
.logo {

  display: flex;

  align-items: center;
```

```
    gap: 1rem;
}
```

```
.logo-icon {
    width: 40px;
    height: 40px;
    background: linear-gradient(135deg, var(--primary-color), var(--secondary-color));
    border-radius: 8px;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 1.5rem;
}
```

```
.logo-text h1 {
    font-size: 1.5rem;
    font-weight: 700;
    background: linear-gradient(135deg, #60a5fa, #3b82f6);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}
```

```
.logo-text p {
    font-size: 0.85rem;
    color: var(--text-secondary);
}
```

```
.header-actions {
```

```
display: flex;
gap: 1rem;
align-items: center;
}
```

```
.status-badge {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  padding: 0.5rem 1rem;
  background: rgba(16, 185, 129, 0.1);
  border: 1px solid var(--success-color);
  border-radius: 20px;
  font-size: 0.85rem;
}
```

```
.status-dot {
  width: 8px;
  height: 8px;
  background: var(--success-color);
  border-radius: 50%;
  animation: pulse 2s infinite;
}
```

```
@keyframes pulse {
  0%, 100% { opacity: 1; }
  50% { opacity: 0.5; }
}
```

```
/* Main Container */
```

```
.container {  
  max-width: 1400px;  
  margin: 0 auto;  
  padding: 2rem;  
}
```

```
/* Metrics Grid */
```

```
.metrics-grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));  
  gap: 1.5rem;  
  margin-bottom: 2rem;  
}
```

```
.metric-card {  
  background: var(--card-bg);  
  border: 1px solid var(--border-color);  
  border-radius: 12px;  
  padding: 1.5rem;  
  transition: transform 0.2s, box-shadow 0.2s;  
}
```

```
.metric-card:hover {  
  transform: translateY(-2px);  
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);  
}
```

```
.metric-header {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  margin-bottom: 1rem;  
}
```

```
.metric-icon {  
  width: 48px;  
  height: 48px;  
  border-radius: 10px;  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  font-size: 1.5rem;  
}
```

```
.metric-icon.blue { background: rgba(37, 99, 235, 0.2); }  
.metric-icon.green { background: rgba(16, 185, 129, 0.2); }  
.metric-icon.orange { background: rgba(245, 158, 11, 0.2); }  
.metric-icon.purple { background: rgba(139, 92, 246, 0.2); }
```

```
.metric-label {  
  color: var(--text-secondary);  
  font-size: 0.9rem;  
  margin-bottom: 0.5rem;  
}
```

```
.metric-value {  
  font-size: 2rem;  
  font-weight: 700;  
  margin-bottom: 0.5rem;  
}
```

```
.metric-change {  
  font-size: 0.85rem;  
  display: flex;  
  align-items: center;  
  gap: 0.25rem;  
}
```

```
.metric-change.positive { color: var(--success-color); }  
.metric-change.negative { color: var(--danger-color); }
```

```
/* Charts Section */
```

```
.charts-section {  
  display: grid;  
  grid-template-columns: 2fr 1fr;  
  gap: 1.5rem;  
  margin-bottom: 2rem;  
}
```

```
.chart-card {  
  background: var(--card-bg);  
  border: 1px solid var(--border-color);
```

```
border-radius: 12px;  
padding: 1.5rem;  
}
```

```
.chart-header {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  margin-bottom: 1.5rem;  
}
```

```
.chart-title {  
  font-size: 1.2rem;  
  font-weight: 600;  
}
```

```
.chart-container {  
  height: 300px;  
  position: relative;  
}
```

```
/* Simple Bar Chart */
```

```
.bar-chart {  
  display: flex;  
  align-items: flex-end;  
  justify-content: space-around;  
  height: 100%;  
  padding: 1rem 0;
```

```
}
```

```
.bar-item {  
  display: flex;  
  flex-direction: column;  
  align-items: center;  
  gap: 0.5rem;  
  flex: 1;  
}
```

```
.bar {  
  width: 40px;  
  background: linear-gradient(180deg, var(--primary-color), var(--secondary-color));  
  border-radius: 4px 4px 0 0;  
  transition: height 0.3s ease;  
  position: relative;  
}
```

```
.bar:hover {  
  opacity: 0.8;  
}
```

```
.bar-label {  
  font-size: 0.75rem;  
  color: var(--text-secondary);  
  text-align: center;  
}
```

```
.bar-value {  
    position: absolute;  
    top: -25px;  
    left: 50%;  
    transform: translateX(-50%);  
    font-size: 0.75rem;  
    font-weight: 600;  
}
```

```
/* Line Chart Simulation */
```

```
.line-chart {  
    position: relative;  
    height: 100%;  
    overflow: hidden;  
}
```

```
.line-svg {  
    width: 100%;  
    height: 100%;  
}
```

```
/* Sources Table */
```

```
.sources-section {  
    background: var(--card-bg);  
    border: 1px solid var(--border-color);  
    border-radius: 12px;  
    padding: 1.5rem;  
    margin-bottom: 2rem;
```

```
}
```

```
.section-header {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  margin-bottom: 1.5rem;  
}
```

```
.btn {  
  padding: 0.75rem 1.5rem;  
  border: none;  
  border-radius: 8px;  
  font-size: 0.9rem;  
  font-weight: 600;  
  cursor: pointer;  
  transition: all 0.2s;  
  display: inline-flex;  
  align-items: center;  
  gap: 0.5rem;  
}
```

```
.btn-primary {  
  background: var(--primary-color);  
  color: white;  
}
```

```
.btn-primary:hover {
```

```
    background: var(--secondary-color);  
    transform: translateY(-1px);  
}
```

```
.btn-success {  
    background: var(--success-color);  
    color: white;  
}
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
}
```

```
th, td {  
    padding: 1rem;  
    text-align: left;  
    border-bottom: 1px solid var(--border-color);  
}
```

```
th {  
    color: var(--text-secondary);  
    font-weight: 600;  
    font-size: 0.85rem;  
    text-transform: uppercase;  
}
```

```
tr:hover {
```

```
    background: rgba(255, 255, 255, 0.02);  
}
```

```
.source-type {  
    display: inline-flex;  
    align-items: center;  
    gap: 0.5rem;  
    padding: 0.25rem 0.75rem;  
    border-radius: 12px;  
    font-size: 0.85rem;  
    font-weight: 500;  
}
```

```
.source-type.iot { background: rgba(37, 99, 235, 0.2); color: #60a5fa; }  
.source-type.api { background: rgba(16, 185, 129, 0.2); color: #34d399; }  
.source-type.db { background: rgba(245, 158, 11, 0.2); color: #fbbf24; }  
.source-type.mq { background: rgba(139, 92, 246, 0.2); color: #a78bfa; }
```

```
.status-indicator {  
    display: inline-flex;  
    align-items: center;  
    gap: 0.5rem;  
}
```

```
.status-dot-small {  
    width: 6px;  
    height: 6px;  
    border-radius: 50%;
```

```
}
```

```
.status-dot-small.active { background: var(--success-color); }
```

```
.status-dot-small.inactive { background: var(--danger-color); }
```

```
/* Jobs Section */
```

```
.jobs-grid {
```

```
  display: grid;
```

```
  grid-template-columns: repeat(auto-fill, minmax(350px, 1fr));
```

```
  gap: 1.5rem;
```

```
  margin-bottom: 2rem;
```

```
}
```

```
.job-card {
```

```
  background: var(--card-bg);
```

```
  border: 1px solid var(--border-color);
```

```
  border-radius: 12px;
```

```
  padding: 1.5rem;
```

```
  transition: all 0.2s;
```

```
}
```

```
.job-card:hover {
```

```
  border-color: var(--primary-color);
```

```
}
```

```
.job-header {
```

```
  display: flex;
```

```
  justify-content: space-between;
```

```
    align-items: start;
    margin-bottom: 1rem;
}
```

```
.job-id {
    font-family: 'Courier New', monospace;
    font-size: 0.85rem;
    color: var(--text-secondary);
}
```

```
.job-status {
    padding: 0.25rem 0.75rem;
    border-radius: 12px;
    font-size: 0.75rem;
    font-weight: 600;
    text-transform: uppercase;
}
```

```
.job-status.running { background: rgba(37, 99, 235, 0.2); color: #60a5fa; }
.job-status.completed { background: rgba(16, 185, 129, 0.2); color: #34d399; }
.job-status.failed { background: rgba(239, 68, 68, 0.2); color: #f87171; }
```

```
.job-stats {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 1rem;
    margin-top: 1rem;
}
```

```
.job-stat {  
    text-align: center;  
}
```

```
.job-stat-value {  
    font-size: 1.5rem;  
    font-weight: 700;  
    color: var(--primary-color);  
}
```

```
.job-stat-label {  
    font-size: 0.75rem;  
    color: var(--text-secondary);  
    margin-top: 0.25rem;  
}
```

```
/* Real-time Log */
```

```
.log-section {  
    background: var(--card-bg);  
    border: 1px solid var(--border-color);  
    border-radius: 12px;  
    padding: 1.5rem;  
}
```

```
.log-container {  
    background: #0a0f1a;  
    border-radius: 8px;
```

```
padding: 1rem;

height: 300px;

overflow-y: auto;

font-family: 'Courier New', monospace;

font-size: 0.85rem;
}

.log-entry {
  padding: 0.5rem 0;

  border-bottom: 1px solid rgba(255, 255, 255, 0.05);

  display: flex;

  gap: 1rem;
}

.log-time {
  color: var(--text-secondary);

  min-width: 80px;
}

.log-level {
  min-width: 60px;

  font-weight: 600;
}

.log-level.info { color: #60a5fa; }
.log-level.success { color: #34d399; }
.log-level.warning { color: #fbbf24; }
.log-level.error { color: #f87171; }
```

```
.log-message {  
  color: var(--text-primary);  
}
```

```
/* Modal */
```

```
.modal {  
  display: none;  
  position: fixed;  
  top: 0;  
  left: 0;  
  width: 100%;  
  height: 100%;  
  background: rgba(0, 0, 0, 0.8);  
  z-index: 1000;  
  align-items: center;  
  justify-content: center;  
}
```

```
.modal.active {  
  display: flex;  
}
```

```
.modal-content {  
  background: var(--card-bg);  
  border: 1px solid var(--border-color);  
  border-radius: 12px;  
  padding: 2rem;
```

```
    max-width: 600px;
    width: 90%;
    max-height: 90vh;
    overflow-y: auto;
}
```

```
.modal-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1.5rem;
}
```

```
.modal-title {
    font-size: 1.5rem;
    font-weight: 600;
}
```

```
.close-btn {
    background: none;
    border: none;
    color: var(--text-secondary);
    font-size: 1.5rem;
    cursor: pointer;
    padding: 0.5rem;
}
```

```
.close-btn:hover {
```

```
    color: var(--text-primary);  
}
```

```
.form-group {  
    margin-bottom: 1.5rem;  
}
```

```
.form-label {  
    display: block;  
    margin-bottom: 0.5rem;  
    color: var(--text-secondary);  
    font-size: 0.9rem;  
}
```

```
.form-input, .form-select {  
    width: 100%;  
    padding: 0.75rem;  
    background: var(--dark-bg);  
    border: 1px solid var(--border-color);  
    border-radius: 8px;  
    color: var(--text-primary);  
    font-size: 0.95rem;  
}
```

```
.form-input:focus, .form-select:focus {  
    outline: none;  
    border-color: var(--primary-color);  
}
```

```
/* Responsive */  
  
@media (max-width: 1024px) {  
  .charts-section {  
    grid-template-columns: 1fr;  
  }  
}
```

```
@media (max-width: 768px) {  
  .header-content {  
    flex-direction: column;  
    gap: 1rem;  
  }
```

```
.metrics-grid {  
  grid-template-columns: 1fr;  
}
```

```
.jobs-grid {  
  grid-template-columns: 1fr;  
}
```

```
table {  
  font-size: 0.85rem;  
}
```

```
th, td {  
  padding: 0.75rem 0.5rem;
```

```
}  
}
```

```
/* Animations */
```

```
@keyframes fadeIn {  
  from { opacity: 0; transform: translateY(10px); }  
  to { opacity: 1; transform: translateY(0); }  
}
```

```
.metric-card, .chart-card, .sources-section, .job-card {  
  animation: fadeIn 0.5s ease-out;  
}
```

```
/* Loading Spinner */
```

```
.spinner {  
  border: 3px solid rgba(255, 255, 255, 0.1);  
  border-top: 3px solid var(--primary-color);  
  border-radius: 50%;  
  width: 40px;  
  height: 40px;  
  animation: spin 1s linear infinite;  
  margin: 2rem auto;  
}
```

```
@keyframes spin {  
  0% { transform: rotate(0deg); }  
  100% { transform: rotate(360deg); }  
}
```

```
</style>

</head>

<body>

  <!-- Header -->

  <header class="header">

    <div class="header-content">

      <div class="logo">

        <div class="logo-icon">  </div>

        <div class="logo-text">

          <h1>DataFlow Pipeline</h1>

          <p>Real-Time Big Data Ingestion System</p>

        </div>

      </div>

      <div class="header-actions">

        <div class="status-badge">

          <div class="status-dot"></div>

          <span>System Operational</span>

        </div>

        <button class="btn btn-primary" onclick="openModal()">+ Add Source</button>

      </div>

    </div>

  </header>

  <!-- Main Container -->

  <div class="container">

    <!-- Metrics Grid -->

    <div class="metrics-grid">
```

```
<div class="metric-card">

  <div class="metric-header">

    <div>

      <div class="metric-label">Active Sources</div>

      <div class="metric-value" id="activeSources">8</div>

      <div class="metric-change positive">↑ 2 new today</div>

    </div>

    <div class="metric-icon blue"> 📶 </div>

  </div>

</div>
```

```
<div class="metric-card">

  <div class="metric-header">

    <div>

      <div class="metric-label">Messages/Second</div>

      <div class="metric-value" id="messagesPerSec">1,547</div>

      <div class="metric-change positive">↑ 12.5%</div>

    </div>

    <div class="metric-icon green"> ⚡ </div>

  </div>

</div>
```

```
<div class="metric-card">

  <div class="metric-header">

    <div>

      <div class="metric-label">Records Today</div>

      <div class="metric-value" id="recordsToday">2.4M</div>

      <div class="metric-change positive">↑ 8.3%</div>

    </div>

  </div>

</div>
```

</div>

<div class="metric-icon orange">  </div>

</div>

</div>

<div class="metric-card">

<div class="metric-header">

<div>

<div class="metric-label">Avg Latency</div>

<div class="metric-value" id="avgLatency">12ms</div>

<div class="metric-change positive">↓ 3ms</div>

</div>

<div class="metric-icon purple">  </div>

</div>

</div>

</div>

<!-- Charts Section -->

<div class="charts-section">

<div class="chart-card">

<div class="chart-header">

<h3 class="chart-title">Ingestion Throughput (Last 24 Hours)</h3>

Real-time

</div>

<div class="chart-container">

<svg class="line-svg" viewBox="0 0 800 300" preserveAspectRatio="none">

<defs>

```

<linearGradient id="lineGradient" x1="0%" y1="0%" x2="0%" y2="100%">
  <stop offset="0%" style="stop-color:#3b82f6;stop-opacity:0.3" />
  <stop offset="100%" style="stop-color:#3b82f6;stop-opacity:0" />
</linearGradient>
</defs>

<path d="M 0 250 Q 50 200 100 180 T 200 150 T 300 120 T 400 140 T 500 100
T 600 80 T 700 90 T 800 60"

fill="none" stroke="#3b82f6" stroke-width="3"/>

<path d="M 0 250 Q 50 200 100 180 T 200 150 T 300 120 T 400 140 T 500 100
T 600 80 T 700 90 T 800 60 L 800 300 L 0 300 Z"

fill="url(#lineGradient)" stroke="none"/>

<!-- Grid lines -->

<line x1="0" y1="250" x2="800" y2="250" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>

<line x1="0" y1="200" x2="800" y2="200" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>

<line x1="0" y1="150" x2="800" y2="150" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>

<line x1="0" y1="100" x2="800" y2="100" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>

<line x1="0" y1="50" x2="800" y2="50" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>

</svg>
</div>
</div>

```

```

<div class="chart-card">

```

```

  <div class="chart-header">

```

```

    <h3 class="chart-title">Records by Source Type</h3>

```

```
</div>

<div class="chart-container">

  <div class="bar-chart">

    <div class="bar-item">

      <div class="bar" style="height: 180px;">

        <div class="bar-value">850K</div>

      </div>

      <div class="bar-label">IoT</div>

    </div>

    <div class="bar-item">

      <div class="bar" style="height: 140px;">

        <div class="bar-value">620K</div>

      </div>

      <div class="bar-label">API</div>

    </div>

    <div class="bar-item">

      <div class="bar" style="height: 200px;">

        <div class="bar-value">920K</div>

      </div>

      <div class="bar-label">Database</div>

    </div>

    <div class="bar-item">

      <div class="bar" style="height: 120px;">

        <div class="bar-value">480K</div>

      </div>

      <div class="bar-label">Queue</div>

    </div>

  </div>

</div>
```

</div>

</div>

</div>

<!-- Active Sources Table -->

<div class="sources-section">

<div class="section-header">

<h3 class="chart-title">Active Data Sources</h3>

<button class="btn btn-primary" onclick="simulateData()">Simulate
Data</button>

</div>

<table>

<thead>

<tr>

<th>Source Name</th>

<th>Type</th>

<th>Status</th>

<th>Records Processed</th>

<th>Throughput</th>

<th>Last Update</th>

</tr>

</thead>

<tbody id="sourcesTable">

<tr>

<td>Warehouse IoT Sensors</td>

<td> 📡 IoT Device</td>

<td>

<div class="status-indicator">

```
<div class="status-dot-small active"></div>

<span>Active</span>

</div>

</td>

<td>856,234</td>

<td>245 rec/s</td>

<td>2 min ago</td>

</tr>

<tr>

<td><strong>User Analytics API</strong></td>

<td><span class="source-type api">🌐 REST API</span></td>

<td>

<div class="status-indicator">

<div class="status-dot-small active"></div>

<span>Active</span>

</div>

</td>

<td>623,891</td>

<td>189 rec/s</td>

<td>1 min ago</td>

</tr>

<tr>

<td><strong>Orders Database CDC</strong></td>

<td><span class="source-type db">💾 Database</span></td>

<td>

<div class="status-indicator">

<div class="status-dot-small active"></div>

<span>Active</span>


```

</td>

```
        <td>234,567</td>

        <td>0 rec/s</td>

        <td>1 hour ago</td>

    </tr>

</tbody>

</table>

</div>

<!-- Active Jobs -->

<h3 style="margin-bottom: 1rem; font-size: 1.3rem;">Active Ingestion Jobs</h3>

<div class="jobs-grid">

    <div class="job-card">

        <div class="job-header">

            <div>

                <div style="font-weight: 600; margin-bottom: 0.25rem;">Warehouse IoT
Stream</div>

                <div class="job-id">job_a3f8b2c1</div>

            </div>

            <span class="job-status running">Running</span>

        </div>

        <div style="color: var(--text-secondary); font-size: 0.9rem; margin-bottom:
1rem;">

            Processing real-time sensor data from 50 devices

        </div>

        <div class="job-stats">

            <div class="job-stat">

                <div class="job-stat-value">856K</div>

                <div class="job-stat-label">Records</div>

            </div>


```

```
<div class="job-stat">
  <div class="job-stat-value">245</div>
  <div class="job-stat-label">Rec/Sec</div>
</div>
</div>
</div>
```

```
<div class="job-card">
  <div class="job-header">
    <div>
      <div style="font-weight: 600; margin-bottom: 0.25rem;">Database CDC
Sync</div>
      <div class="job-id">job_d7e9f4a2</div>
    </div>
    <span class="job-status running">Running</span>
  </div>
  <div style="color: var(--text-secondary); font-size: 0.9rem; margin-bottom:
1rem;">
    Capturing changes from orders database
  </div>
  <div class="job-stats">
    <div class="job-stat">
      <div class="job-stat-value">921K</div>
      <div class="job-stat-label">Records</div>
    </div>
    <div class="job-stat">
      <div class="job-stat-value">312</div>
      <div class="job-stat-label">Rec/Sec</div>
    </div>
  </div>
```

</div>

</div>

<div class="job-card">

<div class="job-header">

<div>

<div style="font-weight: 600; margin-bottom: 0.25rem;">Analytics API
Import</div>

<div class="job-id">job_b5c8d3e6</div>

</div>

Completed

</div>

<div style="color: var(--text-secondary); font-size: 0.9rem; margin-bottom:
1rem;">

Batch import of user analytics data

</div>

<div class="job-stats">

<div class="job-stat">

<div class="job-stat-value">623K</div>

<div class="job-stat-label">Records</div>

</div>

<div class="job-stat">

<div class="job-stat-value">189</div>

<div class="job-stat-label">Rec/Sec</div>

</div>

</div>

</div>

</div>

```
<!-- Real-time Log -->

<div class="log-section">

  <div class="section-header">

    <h3 class="chart-title">Real-time Processing Log</h3>

    <button class="btn" style="background: var(--border-color); color: var(--text-primary);" onclick="clearLog()">Clear</button>

  </div>

  <div class="log-container" id="logContainer">

    <div class="log-entry">

      <span class="log-time">14:32:15</span>

      <span class="log-level info">INFO</span>

      <span class="log-message">Connected to Kafka broker at
localhost:9092</span>

    </div>

    <div class="log-entry">

      <span class="log-time">14:32:16</span>

      <span class="log-level success">SUCCESS</span>

      <span class="log-message">Topic 'ingestion.src_iot_001' created
successfully</span>

    </div>

    <div class="log-entry">

      <span class="log-time">14:32:18</span>

      <span class="log-level info">INFO</span>

      <span class="log-message">Started ingestion job job_a3f8b2c1 for
Warehouse IoT Sensors</span>

    </div>

    <div class="log-entry">

      <span class="log-time">14:32:20</span>

      <span class="log-level success">SUCCESS</span>
```

```
        <span class="log-message">Processed batch of 100 records from IoT
devices</span>
```

```
    </div>
```

```
    <div class="log-entry">
```

```
        <span class="log-time">14:32:22</span>
```

```
        <span class="log-level info">INFO</span>
```

```
        <span class="log-message">Stream processor applied 3 transformation
rules</span>
```

```
    </div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- Modal for Adding Source -->
```

```
<div class="modal" id="addSourceModal">
```

```
    <div class="modal-content">
```

```
        <div class="modal-header">
```

```
            <h2 class="modal-title">Add New Data Source</h2>
```

```
            <button class="close-btn" onclick="closeModal()">&times;</button>
```

```
        </div>
```

```
        <form onsubmit="addSource(event)">
```

```
            <div class="form-group">
```

```
                <label class="form-label">Source Name</label>
```

```
                <input type="text" class="form-input" placeholder="e.g., Production IoT
Sensors" required>
```

```
            </div>
```

```
            <div class="form-group">
```

```
                <label class="form-label">Source Type</label>
```

```
                <select class="form-select" required>
```

```

        <option value="">Select type...</option>
        <option value="iot">IoT Device</option>
        <option value="api">REST API</option>
        <option value="database">Database CDC</option>
        <option value="queue">Message Queue</option>
    </select>
</div>

<div class="form-group">
    <label class="form-label">Endpoint URL</label>

    <input type="text" class="form-input" placeholder="e.g., mqtt://broker:1883"
required>
</div>

<div class="form-group">
    <label class="form-label">Batch Size</label>

    <input type="number" class="form-input" value="100" min="1" max="1000">
</div>

<div class="form-group">
    <label class="form-label">Poll Interval (seconds)</label>

    <input type="number" class="form-input" value="5" min="1" max="60">
</div>

    <button type="submit" class="btn btn-primary" style="width: 100%;">Create
Source</button>
</form>
</div>
</div>

<script>
    // Modal Functions

    function openModal() {

```

```
document.getElementById('addSourceModal').classList.add('active');  
}
```

```
function closeModal() {  
    document.getElementById('addSourceModal').classList.remove('active');  
}
```

```
function addSource(event) {  
    event.preventDefault();  
    alert('Source added successfully! (Demo mode)');  
    closeModal();  
    addLogEntry('success', 'New data source registered and activated');  
}
```

// Log Functions

```
function addLogEntry(level, message) {  
    const logContainer = document.getElementById('logContainer');  
    const time = new Date().toLocaleTimeString('en-US', { hour12: false });  
  
    const entry = document.createElement('div');  
    entry.className = 'log-entry';  
    entry.innerHTML = `  
        <span class="log-time">${time}</span>  
        <span class="log-level ${level}">${level.toUpperCase()}</span>  
        <span class="log-message">${message}</span>  
    `;  
  
    logContainer.appendChild(entry);
```

```
    logContainer.scrollTop = logContainer.scrollHeight;
}
```

```
function clearLog() {
    document.getElementById('logContainer').innerHTML = "";
    addLogEntry('info', 'Log cleared');
}
```

```
// Simulate real-time data updates
```

```
function updateMetrics() {
    // Update messages per second

    const mpsElement = document.getElementById('messagesPerSec');
    let currentMPS = parseInt(mpsElement.textContent.replace(',', ''));
    currentMPS += Math.floor(Math.random() * 50) - 25;
    mpsElement.textContent = currentMPS.toLocaleString();
}
```

```
// Update records today
```

```
const recordsElement = document.getElementById('recordsToday');
let currentRecords = parseFloat(recordsElement.textContent);
currentRecords += 0.01;
recordsElement.textContent = currentRecords.toFixed(1) + 'M';
}
```

```
// Randomly add log entries
```

```
if (Math.random() > 0.7) {
    const messages = [
        'Processed batch of 100 records from IoT devices',
        'Applied transformation rule: normalize_timestamps',
        'Sent 150 messages to Kafka topic ingestion.src_api_001',
    ];
    addLogEntry('info', messages.join(' '));
}
```

```

    'Stream processor stats: 99.8% success rate',
    'Checkpoint saved for job job_d7e9f4a2',
    'New connection established from 192.168.1.45'
  ];
  const levels = ['info', 'success', 'info', 'success'];
  const randomIndex = Math.floor(Math.random() * messages.length);
  addLogEntry(levels[randomIndex], messages[randomIndex]);
}
}

function simulateData() {
  addLogEntry('info', 'Starting data simulation...');

  setTimeout(() => {
    addLogEntry('success', 'Simulated 500 records from IoT devices');
  }, 1000);

  setTimeout(() => {
    addLogEntry('success', 'Simulated 300 records from REST API');
  }, 2000);

  setTimeout(() => {
    addLogEntry('success', 'Simulation complete. Total: 800 records processed');
  }, 3000);
}

// Close modal when clicking outside
window.onclick = function(event) {

```

```
const modal = document.getElementById('addSourceModal');

if (event.target === modal) {
    closeModal();
}
}
```

```
// Start real-time updates

setInterval(updateMetrics, 3000);
```

```
// Initial log entries

setTimeout(() => {
    addLogEntry('info', 'Pipeline monitoring system initialized');
}, 1000);
```

```
setTimeout(() => {
    addLogEntry('success', 'All systems operational');
}, 2000);
```

```
</script>
```

```
</body>
```

```
</html>
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Real-Time Big Data Ingestion Pipeline</title>
```

```
<style>
```

```
* {
```

```
margin: 0;

padding: 0;

box-sizing: border-box;

}
```

```
:root {

  --primary-color: #2563eb;

  --secondary-color: #1e40af;

  --success-color: #10b981;

  --warning-color: #f59e0b;

  --danger-color: #ef4444;

  --dark-bg: #0f172a;

  --card-bg: #1e293b;

  --text-primary: #f1f5f9;

  --text-secondary: #94a3b8;

  --border-color: #334155;

}
```

```
body {

  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;

  background: linear-gradient(135deg, #0f172a 0%, #1e293b 100%);

  color: var(--text-primary);

  min-height: 100vh;

}
```

```
/* Header */

.header {

  background: rgba(30, 41, 59, 0.95);
```

```
    backdrop-filter: blur(10px);  
    padding: 1rem 2rem;  
    border-bottom: 1px solid var(--border-color);  
    position: sticky;  
    top: 0;  
    z-index: 100;  
}
```

```
.header-content {  
    max-width: 1400px;  
    margin: 0 auto;  
    display: flex;  
    justify-content: space-between;  
    align-items: center;  
}
```

```
.logo {  
    display: flex;  
    align-items: center;  
    gap: 1rem;  
}
```

```
.logo-icon {  
    width: 40px;  
    height: 40px;  
    background: linear-gradient(135deg, var(--primary-color), var(--secondary-color));  
    border-radius: 8px;  
    display: flex;
```

```
    align-items: center;
    justify-content: center;
    font-size: 1.5rem;
}
```

```
.logo-text h1 {
    font-size: 1.5rem;
    font-weight: 700;
    background: linear-gradient(135deg, #60a5fa, #3b82f6);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}
```

```
.logo-text p {
    font-size: 0.85rem;
    color: var(--text-secondary);
}
```

```
.header-actions {
    display: flex;
    gap: 1rem;
    align-items: center;
}
```

```
.status-badge {
    display: flex;
    align-items: center;
    gap: 0.5rem;
```

```
padding: 0.5rem 1rem;

background: rgba(16, 185, 129, 0.1);

border: 1px solid var(--success-color);

border-radius: 20px;

font-size: 0.85rem;
}
```

```
.status-dot {

width: 8px;

height: 8px;

background: var(--success-color);

border-radius: 50%;

animation: pulse 2s infinite;
}
```

```
@keyframes pulse {

0%, 100% { opacity: 1; }

50% { opacity: 0.5; }

}
```

```
/* Main Container */

.container {

max-width: 1400px;

margin: 0 auto;

padding: 2rem;

}
```

```
/* Metrics Grid */
```

```
.metrics-grid {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));  
  gap: 1.5rem;  
  margin-bottom: 2rem;  
}
```

```
.metric-card {  
  background: var(--card-bg);  
  border: 1px solid var(--border-color);  
  border-radius: 12px;  
  padding: 1.5rem;  
  transition: transform 0.2s, box-shadow 0.2s;  
}
```

```
.metric-card:hover {  
  transform: translateY(-2px);  
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);  
}
```

```
.metric-header {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  margin-bottom: 1rem;  
}
```

```
.metric-icon {
```

```
width: 48px;
height: 48px;
border-radius: 10px;
display: flex;
align-items: center;
justify-content: center;
font-size: 1.5rem;
}
```

```
.metric-icon.blue { background: rgba(37, 99, 235, 0.2); }
.metric-icon.green { background: rgba(16, 185, 129, 0.2); }
.metric-icon.orange { background: rgba(245, 158, 11, 0.2); }
.metric-icon.purple { background: rgba(139, 92, 246, 0.2); }
```

```
.metric-label {
  color: var(--text-secondary);
  font-size: 0.9rem;
  margin-bottom: 0.5rem;
}
```

```
.metric-value {
  font-size: 2rem;
  font-weight: 700;
  margin-bottom: 0.5rem;
}
```

```
.metric-change {
  font-size: 0.85rem;
```

```
display: flex;
align-items: center;
gap: 0.25rem;
}
```

```
.metric-change.positive { color: var(--success-color); }
.metric-change.negative { color: var(--danger-color); }
```

```
/* Charts Section */
```

```
.charts-section {
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 1.5rem;
  margin-bottom: 2rem;
}
```

```
.chart-card {
  background: var(--card-bg);
  border: 1px solid var(--border-color);
  border-radius: 12px;
  padding: 1.5rem;
}
```

```
.chart-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1.5rem;
```

```
}
```

```
.chart-title {  
    font-size: 1.2rem;  
    font-weight: 600;  
}
```

```
.chart-container {  
    height: 300px;  
    position: relative;  
}
```

```
/* Simple Bar Chart */
```

```
.bar-chart {  
    display: flex;  
    align-items: flex-end;  
    justify-content: space-around;  
    height: 100%;  
    padding: 1rem 0;  
}
```

```
.bar-item {  
    display: flex;  
    flex-direction: column;  
    align-items: center;  
    gap: 0.5rem;  
    flex: 1;  
}
```

```
.bar {  
  width: 40px;  
  background: linear-gradient(180deg, var(--primary-color), var(--secondary-color));  
  border-radius: 4px 4px 0 0;  
  transition: height 0.3s ease;  
  position: relative;  
}
```

```
.bar:hover {  
  opacity: 0.8;  
}
```

```
.bar-label {  
  font-size: 0.75rem;  
  color: var(--text-secondary);  
  text-align: center;  
}
```

```
.bar-value {  
  position: absolute;  
  top: -25px;  
  left: 50%;  
  transform: translateX(-50%);  
  font-size: 0.75rem;  
  font-weight: 600;  
}
```

```
/* Line Chart Simulation */
```

```
.line-chart {  
  position: relative;  
  height: 100%;  
  overflow: hidden;  
}
```

```
.line-svg {  
  width: 100%;  
  height: 100%;  
}
```

```
/* Sources Table */
```

```
.sources-section {  
  background: var(--card-bg);  
  border: 1px solid var(--border-color);  
  border-radius: 12px;  
  padding: 1.5rem;  
  margin-bottom: 2rem;  
}
```

```
.section-header {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
  margin-bottom: 1.5rem;  
}
```

```
.btn {  
  padding: 0.75rem 1.5rem;  
  border: none;  
  border-radius: 8px;  
  font-size: 0.9rem;  
  font-weight: 600;  
  cursor: pointer;  
  transition: all 0.2s;  
  display: inline-flex;  
  align-items: center;  
  gap: 0.5rem;  
}
```

```
.btn-primary {  
  background: var(--primary-color);  
  color: white;  
}
```

```
.btn-primary:hover {  
  background: var(--secondary-color);  
  transform: translateY(-1px);  
}
```

```
.btn-success {  
  background: var(--success-color);  
  color: white;  
}
```

```
table {  
  width: 100%;  
  border-collapse: collapse;  
}
```

```
th, td {  
  padding: 1rem;  
  text-align: left;  
  border-bottom: 1px solid var(--border-color);  
}
```

```
th {  
  color: var(--text-secondary);  
  font-weight: 600;  
  font-size: 0.85rem;  
  text-transform: uppercase;  
}
```

```
tr:hover {  
  background: rgba(255, 255, 255, 0.02);  
}
```

```
.source-type {  
  display: inline-flex;  
  align-items: center;  
  gap: 0.5rem;  
  padding: 0.25rem 0.75rem;  
  border-radius: 12px;
```

```
font-size: 0.85rem;

font-weight: 500;

}
```

```
.source-type.iot { background: rgba(37, 99, 235, 0.2); color: #60a5fa; }
.source-type.api { background: rgba(16, 185, 129, 0.2); color: #34d399; }
.source-type.db { background: rgba(245, 158, 11, 0.2); color: #fbbf24; }
.source-type.mq { background: rgba(139, 92, 246, 0.2); color: #a78bfa; }
```

```
.status-indicator {

  display: inline-flex;

  align-items: center;

  gap: 0.5rem;

}
```

```
.status-dot-small {

  width: 6px;

  height: 6px;

  border-radius: 50%;

}
```

```
.status-dot-small.active { background: var(--success-color); }
.status-dot-small.inactive { background: var(--danger-color); }
```

```
/* Jobs Section */
```

```
.jobs-grid {

  display: grid;

  grid-template-columns: repeat(auto-fill, minmax(350px, 1fr));

}
```

```
    gap: 1.5rem;
    margin-bottom: 2rem;
}
```

```
.job-card {
    background: var(--card-bg);
    border: 1px solid var(--border-color);
    border-radius: 12px;
    padding: 1.5rem;
    transition: all 0.2s;
}
```

```
.job-card:hover {
    border-color: var(--primary-color);
}
```

```
.job-header {
    display: flex;
    justify-content: space-between;
    align-items: start;
    margin-bottom: 1rem;
}
```

```
.job-id {
    font-family: 'Courier New', monospace;
    font-size: 0.85rem;
    color: var(--text-secondary);
}
```

```
.job-status {  
  padding: 0.25rem 0.75rem;  
  border-radius: 12px;  
  font-size: 0.75rem;  
  font-weight: 600;  
  text-transform: uppercase;  
}
```

```
.job-status.running { background: rgba(37, 99, 235, 0.2); color: #60a5fa; }  
.job-status.completed { background: rgba(16, 185, 129, 0.2); color: #34d399; }  
.job-status.failed { background: rgba(239, 68, 68, 0.2); color: #f87171; }
```

```
.job-stats {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
  gap: 1rem;  
  margin-top: 1rem;  
}
```

```
.job-stat {  
  text-align: center;  
}
```

```
.job-stat-value {  
  font-size: 1.5rem;  
  font-weight: 700;  
  color: var(--primary-color);
```

```
}
```

```
.job-stat-label {  
    font-size: 0.75rem;  
    color: var(--text-secondary);  
    margin-top: 0.25rem;  
}
```

```
/* Real-time Log */
```

```
.log-section {  
    background: var(--card-bg);  
    border: 1px solid var(--border-color);  
    border-radius: 12px;  
    padding: 1.5rem;  
}
```

```
.log-container {  
    background: #0a0f1a;  
    border-radius: 8px;  
    padding: 1rem;  
    height: 300px;  
    overflow-y: auto;  
    font-family: 'Courier New', monospace;  
    font-size: 0.85rem;  
}
```

```
.log-entry {  
    padding: 0.5rem 0;
```

```
border-bottom: 1px solid rgba(255, 255, 255, 0.05);  
display: flex;  
gap: 1rem;  
}
```

```
.log-time {  
  color: var(--text-secondary);  
  min-width: 80px;  
}
```

```
.log-level {  
  min-width: 60px;  
  font-weight: 600;  
}
```

```
.log-level.info { color: #60a5fa; }  
.log-level.success { color: #34d399; }  
.log-level.warning { color: #fbbf24; }  
.log-level.error { color: #f87171; }
```

```
.log-message {  
  color: var(--text-primary);  
}
```

```
/* Modal */
```

```
.modal {  
  display: none;  
  position: fixed;
```

```
top: 0;
left: 0;
width: 100%;
height: 100%;
background: rgba(0, 0, 0, 0.8);
z-index: 1000;
align-items: center;
justify-content: center;
}
```

```
.modal.active {
  display: flex;
}
```

```
.modal-content {
  background: var(--card-bg);
  border: 1px solid var(--border-color);
  border-radius: 12px;
  padding: 2rem;
  max-width: 600px;
  width: 90%;
  max-height: 90vh;
  overflow-y: auto;
}
```

```
.modal-header {
  display: flex;
  justify-content: space-between;
```

```
    align-items: center;
    margin-bottom: 1.5rem;
}
```

```
.modal-title {
    font-size: 1.5rem;
    font-weight: 600;
}
```

```
.close-btn {
    background: none;
    border: none;
    color: var(--text-secondary);
    font-size: 1.5rem;
    cursor: pointer;
    padding: 0.5rem;
}
```

```
.close-btn:hover {
    color: var(--text-primary);
}
```

```
.form-group {
    margin-bottom: 1.5rem;
}
```

```
.form-label {
    display: block;
```

```
margin-bottom: 0.5rem;

color: var(--text-secondary);

font-size: 0.9rem;
}
```

```
.form-input, .form-select {

width: 100%;

padding: 0.75rem;

background: var(--dark-bg);

border: 1px solid var(--border-color);

border-radius: 8px;

color: var(--text-primary);

font-size: 0.95rem;
}
```

```
.form-input:focus, .form-select:focus {

outline: none;

border-color: var(--primary-color);
}
```

```
/* Responsive */

@media (max-width: 1024px) {

charts-section {

grid-template-columns: 1fr;

}

}
```

```
@media (max-width: 768px) {
```

```
.header-content {
    flex-direction: column;
    gap: 1rem;
}

.metrics-grid {
    grid-template-columns: 1fr;
}

.jobs-grid {
    grid-template-columns: 1fr;
}

table {
    font-size: 0.85rem;
}

th, td {
    padding: 0.75rem 0.5rem;
}

}

/* Animations */
@keyframes fadeIn {
    from { opacity: 0; transform: translateY(10px); }
    to { opacity: 1; transform: translateY(0); }
}
```

```

.metric-card, .chart-card, .sources-section, .job-card {
  animation: fadeIn 0.5s ease-out;
}

/* Loading Spinner */
.spinner {
  border: 3px solid rgba(255, 255, 255, 0.1);
  border-top: 3px solid var(--primary-color);
  border-radius: 50%;
  width: 40px;
  height: 40px;
  animation: spin 1s linear infinite;
  margin: 2rem auto;
}

@keyframes spin {
  0% { transform: rotate(0deg); }
  100% { transform: rotate(360deg); }
}

</style>
</head>
<body>
  <!-- Header -->
  <header class="header">
    <div class="header-content">
      <div class="logo">
        <div class="logo-icon">  </div>
        <div class="logo-text">

```

```
<h1>DataFlow Pipeline</h1>

<p>Real-Time Big Data Ingestion System</p>

</div>

</div>

<div class="header-actions">

  <div class="status-badge">

    <div class="status-dot"></div>

    <span>System Operational</span>

  </div>

  <button class="btn btn-primary" onclick="openModal()">+ Add
Source</button>

</div>

</div>

</header>
```

```
<!-- Main Container -->

<div class="container">

  <!-- Metrics Grid -->

  <div class="metrics-grid">

    <div class="metric-card">

      <div class="metric-header">

        <div>

          <div class="metric-label">Active Sources</div>

          <div class="metric-value" id="activeSources">8</div>

          <div class="metric-change positive">↑ 2 new today</div>

        </div>

        <div class="metric-icon blue"> 📡 </div>

      </div>

    </div>

  </div>

</div>
```

</div>

<div class="metric-card">

<div class="metric-header">

<div>

<div class="metric-label">Messages/Second</div>

<div class="metric-value" id="messagesPerSec">1,547</div>

<div class="metric-change positive">↑ 12.5%</div>

</div>

<div class="metric-icon green"> ⚡ </div>

</div>

</div>

<div class="metric-card">

<div class="metric-header">

<div>

<div class="metric-label">Records Today</div>

<div class="metric-value" id="recordsToday">2.4M</div>

<div class="metric-change positive">↑ 8.3%</div>

</div>

<div class="metric-icon orange"> 📊 </div>

</div>

</div>

<div class="metric-card">

<div class="metric-header">

<div>

<div class="metric-label">Avg Latency</div>

```

        <div class="metric-value" id="avgLatency">12ms</div>

        <div class="metric-change positive">↓ 3ms</div>

    </div>

    <div class="metric-icon purple"> 🕒 </div>

</div>

</div>

<!-- Charts Section -->

<div class="charts-section">

    <div class="chart-card">

        <div class="chart-header">

            <h3 class="chart-title">Ingestion Throughput (Last 24 Hours)</h3>

            <span style="color: var(--text-secondary); font-size: 0.85rem;">Real-
time</span>

        </div>

        <div class="chart-container">

            <svg class="line-svg" viewBox="0 0 800 300" preserveAspectRatio="none">

                <defs>

                    <linearGradient id="lineGradient" x1="0%" y1="0%" x2="0%" y2="100%">

                        <stop offset="0%" style="stop-color:#3b82f6;stop-opacity:0.3" />

                        <stop offset="100%" style="stop-color:#3b82f6;stop-opacity:0" />

                    </linearGradient>

                </defs>

                <path d="M 0 250 Q 50 200 100 180 T 200 150 T 300 120 T 400 140 T 500 100
T 600 80 T 700 90 T 800 60"

                    fill="none" stroke="#3b82f6" stroke-width="3"/>

                <path d="M 0 250 Q 50 200 100 180 T 200 150 T 300 120 T 400 140 T 500 100
T 600 80 T 700 90 T 800 60 L 800 300 L 0 300 Z"

```

```
fill="url(#lineGradient)" stroke="none"/>
```

```
<!-- Grid lines -->
```

```
<line x1="0" y1="250" x2="800" y2="250" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>
```

```
<line x1="0" y1="200" x2="800" y2="200" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>
```

```
<line x1="0" y1="150" x2="800" y2="150" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>
```

```
<line x1="0" y1="100" x2="800" y2="100" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>
```

```
<line x1="0" y1="50" x2="800" y2="50" stroke="#334155" stroke-width="1"
stroke-dasharray="5,5"/>
```

```
</svg>
```

```
</div>
```

```
</div>
```

```
<div class="chart-card">
```

```
<div class="chart-header">
```

```
<h3 class="chart-title">Records by Source Type</h3>
```

```
</div>
```

```
<div class="chart-container">
```

```
<div class="bar-chart">
```

```
<div class="bar-item">
```

```
<div class="bar" style="height: 180px;">
```

```
<div class="bar-value">850K</div>
```

```
</div>
```

```
<div class="bar-label">IoT</div>
```

```
</div>
```

```
<div class="bar-item">

  <div class="bar" style="height: 140px;">

    <div class="bar-value">620K</div>

  </div>

  <div class="bar-label">API</div>

</div>

<div class="bar-item">

  <div class="bar" style="height: 200px;">

    <div class="bar-value">920K</div>

  </div>

  <div class="bar-label">Database</div>

</div>

<div class="bar-item">

  <div class="bar" style="height: 120px;">

    <div class="bar-value">480K</div>

  </div>

  <div class="bar-label">Queue</div>

</div>

</div>

</div>

</div>

</div>

<!-- Active Sources Table -->

<div class="sources-section">

  <div class="section-header">

    <h3 class="chart-title">Active Data Sources</h3>
```

```
        <button class="btn btn-primary" onclick="simulateData()">Simulate
Data</button>
```

```
    </div>
```

```
    <table>
```

```
        <thead>
```

```
            <tr>
```

```
                <th>Source Name</th>
```

```
                <th>Type</th>
```

```
                <th>Status</th>
```

```
                <th>Records Processed</th>
```

```
                <th>Throughput</th>
```

```
                <th>Last Update</th>
```

```
            </tr>
```

```
        </thead>
```

```
        <tbody id="sourcesTable">
```

```
            <tr>
```

```
                <td><strong>Warehouse IoT Sensors</strong></td>
```

```
                <td><span class="source-type iot"> 📡 IoT Device</span></td>
```

```
                <td>
```

```
                    <div class="status-indicator">
```

```
                        <div class="status-dot-small active"></div>
```

```
                        <span>Active</span>
```

```
                    </div>
```

```
                </td>
```

```
                <td>856,234</td>
```

```
                <td>245 rec/s</td>
```

```
                <td>2 min ago</td>
```

```
            </tr>
```

```
<tr>

  <td><strong>User Analytics API</strong></td>

  <td><span class="source-type api">  REST API</span></td>

  <td>

    <div class="status-indicator">

      <div class="status-dot-small active"></div>

      <span>Active</span>

    </div>

  </td>

  <td>623,891</td>

  <td>189 rec/s</td>

  <td>1 min ago</td>

</tr>

<tr>

  <td><strong>Orders Database CDC</strong></td>

  <td><span class="source-type db">  Database</span></td>

  <td>

    <div class="status-indicator">

      <div class="status-dot-small active"></div>

      <span>Active</span>

    </div>

  </td>

  <td>921,456</td>

  <td>312 rec/s</td>

  <td>Just now</td>

</tr>

<tr>

  <td><strong>Notification Queue</strong></td>
```

```
<td><span class="source-type mq">✉ Message Queue</span></td>
<td>
  <div class="status-indicator">
    <div class="status-dot-small active"></div>
    <span>Active</span>
  </div>
</td>
<td>478,923</td>
<td>156 rec/s</td>
<td>3 min ago</td>
</tr>
<tr>
<td><strong>Payment Gateway</strong></td>
<td><span class="source-type api">🌐 REST API</span></td>
<td>
  <div class="status-indicator">
    <div class="status-dot-small inactive"></div>
    <span>Inactive</span>
  </div>
</td>
<td>234,567</td>
<td>0 rec/s</td>
<td>1 hour ago</td>
</tr>
</tbody>
</table>
</div>
```

```
<!-- Active Jobs -->

<h3 style="margin-bottom: 1rem; font-size: 1.3rem;">Active Ingestion Jobs</h3>

<div class="jobs-grid">

  <div class="job-card">

    <div class="job-header">

      <div>

        <div style="font-weight: 600; margin-bottom: 0.25rem;">Warehouse IoT
Stream</div>

        <div class="job-id">job_a3f8b2c1</div>

      </div>

      <span class="job-status running">Running</span>

    </div>

    <div style="color: var(--text-secondary); font-size: 0.9rem; margin-bottom:
1rem;">

      Processing real-time sensor data from 50 devices

    </div>

    <div class="job-stats">

      <div class="job-stat">

        <div class="job-stat-value">856K</div>

        <div class="job-stat-label">Records</div>

      </div>

      <div class="job-stat">

        <div class="job-stat-value">245</div>

        <div class="job-stat-label">Rec/Sec</div>

      </div>

    </div>

  </div>

  <div class="job-card">
```

```
<div class="job-header">

  <div>

    <div style="font-weight: 600; margin-bottom: 0.25rem;">Database CDC
Sync</div>

    <div class="job-id">job_d7e9f4a2</div>

  </div>

  <span class="job-status running">Running</span>

</div>

<div style="color: var(--text-secondary); font-size: 0.9rem; margin-bottom:
1rem;">

  Capturing changes from orders database

</div>

<div class="job-stats">

  <div class="job-stat">

    <div class="job-stat-value">921K</div>

    <div class="job-stat-label">Records</div>

  </div>

  <div class="job-stat">

    <div class="job-stat-value">312</div>

    <div class="job-stat-label">Rec/Sec</div>

  </div>

</div>

</div>

<div class="job-card">

  <div class="job-header">

    <div>

      <div style="font-weight: 600; margin-bottom: 0.25rem;">Analytics API
Import</div>
```

```
        <div class="job-id">job_b5c8d3e6</div>

    </div>

    <span class="job-status completed">Completed</span>

</div>

<div style="color: var(--text-secondary); font-size: 0.9rem; margin-bottom: 1rem;">

    Batch import of user analytics data

</div>

<div class="job-stats">

    <div class="job-stat">

        <div class="job-stat-value">623K</div>

        <div class="job-stat-label">Records</div>

    </div>

    <div class="job-stat">

        <div class="job-stat-value">189</div>

        <div class="job-stat-label">Rec/Sec</div>

    </div>

</div>

</div>

</div>

<!-- Real-time Log -->

<div class="log-section">

    <div class="section-header">

        <h3 class="chart-title">Real-time Processing Log</h3>

        <button class="btn" style="background: var(--border-color); color: var(--text-primary);" onclick="clearLog()">Clear</button>

    </div>

    <div class="log-container" id="logContainer">
```

```
<div class="log-entry">

  <span class="log-time">14:32:15</span>

  <span class="log-level info">INFO</span>

  <span class="log-message">Connected to Kafka broker at
localhost:9092</span>

</div>

<div class="log-entry">

  <span class="log-time">14:32:16</span>

  <span class="log-level success">SUCCESS</span>

  <span class="log-message">Topic 'ingestion.src_iot_001' created
successfully</span>

</div>

<div class="log-entry">

  <span class="log-time">14:32:18</span>

  <span class="log-level info">INFO</span>

  <span class="log-message">Started ingestion job job_a3f8b2c1 for
Warehouse IoT Sensors</span>

</div>

<div class="log-entry">

  <span class="log-time">14:32:20</span>

  <span class="log-level success">SUCCESS</span>

  <span class="log-message">Processed batch of 100 records from IoT
devices</span>

</div>

<div class="log-entry">

  <span class="log-time">14:32:22</span>

  <span class="log-level info">INFO</span>

  <span class="log-message">Stream processor applied 3 transformation
rules</span>
```

</div>

</div>

</div>

</div>

<!-- Modal for Adding Source -->

<div class="modal" id="addSourceModal">

<div class="modal-content">

<div class="modal-header">

<h2 class="modal-title">Add New Data Source</h2>

<button class="close-btn" onclick="closeModal()">×</button>

</div>

<form onsubmit="addSource(event)">

<div class="form-group">

<label class="form-label">Source Name</label>

<input type="text" class="form-input" placeholder="e.g., Production IoT
Sensors" required>

</div>

<div class="form-group">

<label class="form-label">Source Type</label>

<select class="form-select" required>

<option value="">Select type...</option>

<option value="iot">IoT Device</option>

<option value="api">REST API</option>

<option value="database">Database CDC</option>

<option value="queue">Message Queue</option>

</select>

</div>

```
<div class="form-group">

  <label class="form-label">Endpoint URL</label>

  <input type="text" class="form-input" placeholder="e.g., mqtt://broker:1883"
required>

</div>

<div class="form-group">

  <label class="form-label">Batch Size</label>

  <input type="number" class="form-input" value="100" min="1" max="1000">

</div>

<div class="form-group">

  <label class="form-label">Poll Interval (seconds)</label>

  <input type="number" class="form-input" value="5" min="1" max="60">

</div>

<button type="submit" class="btn btn-primary" style="width: 100%;">Create
Source</button>

</form>

</div>

</div>
```

```
<script>

  // Modal Functions

  function openModal() {

    document.getElementById('addSourceModal').classList.add('active');

  }

  function closeModal() {

    document.getElementById('addSourceModal').classList.remove('active');

  }
```

```
function addSource(event) {  
    event.preventDefault();  
    alert('Source added successfully! (Demo mode)');  
    closeModal();  
    addLogEntry('success', 'New data source registered and activated');  
}
```

// Log Functions

```
function addLogEntry(level, message) {  
    const logContainer = document.getElementById('logContainer');  
    const time = new Date().toLocaleTimeString('en-US', { hour12: false });  
  
    const entry = document.createElement('div');  
    entry.className = 'log-entry';  
    entry.innerHTML = `  
        <span class="log-time">${time}</span>  
        <span class="log-level ${level}">${level.toUpperCase()}</span>  
        <span class="log-message">${message}</span>  
    `;  
  
    logContainer.appendChild(entry);  
    logContainer.scrollTop = logContainer.scrollHeight;  
}
```

```
function clearLog() {  
    document.getElementById('logContainer').innerHTML = "";  
    addLogEntry('info', 'Log cleared');  
}
```

```
// Simulate real-time data updates

function updateMetrics() {

  // Update messages per second

  const mpsElement = document.getElementById('messagesPerSec');

  let currentMPS = parseInt(mpsElement.textContent.replace(',', ''));

  currentMPS += Math.floor(Math.random() * 50) - 25;

  mpsElement.textContent = currentMPS.toLocaleString();


  // Update records today

  const recordsElement = document.getElementById('recordsToday');

  let currentRecords = parseFloat(recordsElement.textContent);

  currentRecords += 0.01;

  recordsElement.textContent = currentRecords.toFixed(1) + 'M';


  // Randomly add log entries

  if (Math.random() > 0.7) {

    const messages = [

      'Processed batch of 100 records from IoT devices',

      'Applied transformation rule: normalize_timestamps',

      'Sent 150 messages to Kafka topic ingestion.src_api_001',

      'Stream processor stats: 99.8% success rate',

      'Checkpoint saved for job job_d7e9f4a2',

      'New connection established from 192.168.1.45'

    ];

    const levels = ['info', 'success', 'info', 'success'];

    const randomIndex = Math.floor(Math.random() * messages.length);

    addLogEntry(levels[randomIndex], messages[randomIndex]);

  }

}
```

```
}  
}
```

```
function simulateData() {  
  addLogEntry('info', 'Starting data simulation...');  
  
  setTimeout(() => {  
    addLogEntry('success', 'Simulated 500 records from IoT devices');  
  }, 1000);  
  
  setTimeout(() => {  
    addLogEntry('success', 'Simulated 300 records from REST API');  
  }, 2000);  
  
  setTimeout(() => {  
    addLogEntry('success', 'Simulation complete. Total: 800 records processed');  
  }, 3000);  
}
```

```
// Close modal when clicking outside  
window.onclick = function(event) {  
  const modal = document.getElementById('addSourceModal');  
  if (event.target === modal) {  
    closeModal();  
  }  
}
```

```
// Start real-time updates
```

```
setInterval(updateMetrics, 3000);
```

```
// Initial log entries
```

```
setTimeout(() => {
```

```
    addLogEntry('info', 'Pipeline monitoring system initialized');
```

```
}, 1000);
```

```
setTimeout(() => {
```

```
    addLogEntry('success', 'All systems operational');
```

```
}, 2000);
```

```
</script>
```

```
</body>
```

```
</html>
```